

String functions

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DBMS LAB 3-SQL Functions

1.STRING FUNCTIONS:

1.UPPER()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES  
(101, ' john ', 'ABC@exp.com', '2023-01-15', 65000),  
(102, ' smith', 'jNC@exp.com', '2022-05-20', 78000),  
(103, ' allision ', 'MNC@exp.com', '2021-11-10', 82000),  
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT  
    emp_id,  
    UPPER(emp_name) AS upper_name  
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 65000  |
| 102    | smith     | jNC@exp.com | 2022-05-20   | 78000  |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 82000  |
| 104    | chandler  | YUO@exp.com | 2023-08-01   | 55000  |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 90000  |
+-----+-----+-----+-----+-----+

+-----+-----+
| emp_id | upper_name |
+-----+-----+
| 101    | JOHN       |
| 102    | SMITH      |
| 103    | ALLISION   |
| 104    | CHANDLER   |
| 105    | CHARLIE    |
+-----+-----+
```

2.LOWER()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES  
(101, ' john ', 'ABC@exp.com', '2023-01-15', 65000),  
(102, ' smith', 'jNC@exp.com', '2022-05-20', 78000),  
(103, ' allision ', 'MNC@exp.com', '2021-11-10', 82000),  
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT  
    emp_id,  
    LOWER(email) AS lower_email  
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 65000  |
| 102    | smith     | jnc@exp.com | 2022-05-20   | 78000  |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 82000  |
| 104    | chandler  | YUO@exp.com | 2023-08-01   | 55000  |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 90000  |
+-----+-----+-----+-----+-----+

+-----+-----+
| emp_id | lower_email |
+-----+-----+
| 101    | abc@exp.com |
| 102    | jnc@exp.com |
| 103    | mnc@exp.com |
| 104    | yuo@exp.com |
| 105    | ghe@exp.com |
+-----+-----+
```

3.LENGTH()-

```
CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  email VARCHAR(100),
  joining_date DATE,
  salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15', 65000),
(102, ' smith', 'jnc@exp.com', '2022-05-20', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10', 82000),
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
  emp_id,
  emp_name,
  LENGTH(emp_name) AS name_length
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 65000  |
| 102    | smith     | jNC@exp.com | 2022-05-20   | 78000  |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 82000  |
| 104    | chandler  | YUO@exp.com | 2023-08-01   | 55000  |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 90000  |
+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | emp_name  | name_length |
+-----+-----+-----+
| 101    | john      | 9           |
| 102    | smith     | 6           |
| 103    | allision  | 12          |
| 104    | chandler  | 8           |
| 105    | charlie   | 7           |
+-----+-----+-----+
```

4.CONCAT()-

```
CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  email VARCHAR(100),
  joining_date DATE,
  salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15', 65000),
(102, ' smith', 'jNC@exp.com', '2022-05-20', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10', 82000),
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
  emp_id,
  CONCAT(emp_name, ' (Email: ', email, ' - Salary: $', salary, ')') AS employee_summary
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 65000  |
| 102    | smith     | jnc@exp.com | 2022-05-20   | 78000  |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 82000  |
| 104    | chandler  | YUO@exp.com | 2023-08-01   | 55000  |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 90000  |
+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+
| emp_id | employee_summary |
+-----+-----+-----+-----+-----+
| 101    | john (Email: ABC@exp.com - Salary: $65000) |
| 102    | smith (Email: jnc@exp.com - Salary: $78000) |
| 103    | allision (Email: MNC@exp.com - Salary: $82000) |
| 104    | chandler (Email: YUO@exp.com - Salary: $55000) |
| 105    | charlie (Email: GHe@exp.com - Salary: $90000) |
+-----+-----+-----+-----+-----+
```

5.SUBSTRING()-

```
CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  email VARCHAR(100),
  joining_date DATE,
  salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15', 65000),
(102, ' smith', 'jnc@exp.com', '2022-05-20', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10', 82000),
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
  emp_id,
  email,
  SUBSTRING(email, 1, 3) AS email_prefix
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 65000  |
| 102    | smith     | jNC@exp.com | 2022-05-20   | 78000  |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 82000  |
| 104    | chandler  | YUO@exp.com | 2023-08-01   | 55000  |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 90000  |
+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | email      | email_prefix |
+-----+-----+-----+
| 101    | ABC@exp.com | ABC          |
| 102    | jNC@exp.com | jNC          |
| 103    | MNC@exp.com | MNC          |
| 104    | YUO@exp.com | YUO          |
| 105    | GHe@exp.com | GHe          |
+-----+-----+-----+
```

6.TRIM()-

```
CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  email VARCHAR(100),
  joining_date DATE,
  salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15', 65000),
(102, ' smith', 'jNC@exp.com', '2022-05-20', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10', 82000),
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
  emp_id,
  emp_name AS raw_name,
  TRIM(emp_name) AS cleaned_name
FROM employee;
```


Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 65000  |
| 102    | smith     | jNC@exp.com | 2022-05-20   | 78000  |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 82000  |
| 104    | chandler  | YUO@exp.com | 2023-08-01   | 55000  |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 90000  |
+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | raw_name  | cleaned_name |
+-----+-----+-----+
| 101    | john      | john         |
| 102    | smith     | smith        |
| 103    | allision  | allision     |
| 104    | chandler  | chandler     |
| 105    | charlie   | charlie      |
+-----+-----+-----+
```

Numeric functions

2.NUMERIC FUNCTIONS:

1.ROUND()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    salary INT  
);  
  
INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES  
(101, 'john', 'ABC@exp.com', '2023-01-15', 65000),  
(102, 'smith', 'jnc@exp.com', '2022-05-20', 78000),  
(103, 'allision', 'MNC@exp.com', '2021-11-10', 82000),  
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);  
  
SHOW TABLES;  
  
SELECT * FROM employee;  
  
SELECT  
    emp_id,  
    emp_name,  
    salary,  
    ROUND(salary / 12) AS monthly_salary_rounded  
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 65000 |
| 102    | smith     | jNC@exp.com | 2022-05-20   | 78000 |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 82000 |
| 104    | chandler  | YUC@exp.com | 2023-08-01   | 55000 |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 90000 |
+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+
| emp_id | emp_name  | salary | monthly_salary_rounded |
+-----+-----+-----+-----+
| 101    | john      | 65000 | 5417 |
| 102    | smith     | 78000 | 6500 |
| 103    | allision  | 82000 | 6833 |
| 104    | chandler  | 55000 | 4583 |
| 105    | charlie   | 90000 | 7500 |
+-----+-----+-----+-----+
```

2.CEIL()-

```

CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    email VARCHAR(100),
    joining_date DATE,
    salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES
(101, 'john', 'ABC@exp.com', '2023-01-15', 65000),
(102, 'smith', 'jnc@exp.com', '2022-05-20', 78000),
(103, 'allision', 'MNC@exp.com', '2021-11-10', 82000),
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
    emp_id,
    emp_name,
    (salary / 7) AS exact_daily_rate,
    CEIL(salary / 7) AS ceiled_daily_rate
FROM employee;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john     | ABC@exp.com | 2023-01-15   | 65000 |
| 102    | smith    | jnc@exp.com | 2022-05-20   | 78000 |
| 103    | allision | MNC@exp.com | 2021-11-10   | 82000 |
| 104    | chandler | YUO@exp.com | 2023-08-01   | 55000 |
| 105    | charlie  | GHe@exp.com | 2020-03-12   | 90000 |
+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name | exact_daily_rate | ceiled_daily_rate |
+-----+-----+-----+-----+-----+
| 101    | john     | 9285.7143        | 9286              |
| 102    | smith    | 11142.8571       | 11143             |
| 103    | allision | 11714.2857       | 11715             |
| 104    | chandler | 7857.1429        | 7858              |
| 105    | charlie  | 12857.1429       | 12858             |
+-----+-----+-----+-----+-----+

```

3.FLOOR()-

```

CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  email VARCHAR(100),
  joining_date DATE,
  salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES
(101, 'john', 'ABC@exp.com', '2023-01-15', 65000),
(102, 'smith', 'jnc@exp.com', '2022-05-20', 78000),
(103, 'allision', 'MNC@exp.com', '2021-11-10', 82000),
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);

```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```

SELECT
  emp_id,
  emp_name,
  (salary / 7) AS exact_daily_rate,
  FLOOR(salary / 7) AS floored_daily_rate
FROM employee;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | salary |
+-----+-----+-----+-----+-----+
| 101    | john     | ABC@exp.com | 2023-01-15   | 65000  |
| 102    | smith    | jnc@exp.com | 2022-05-20   | 78000  |
| 103    | allision | MNC@exp.com | 2021-11-10   | 82000  |
| 104    | chandler | YUO@exp.com | 2023-08-01   | 55000  |
| 105    | charlie  | GHe@exp.com | 2020-03-12   | 90000  |
+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+
| emp_id | emp_name | exact_daily_rate | floored_daily_rate |
+-----+-----+-----+-----+-----+
| 101    | john     | 9285.7143        | 9285               |
| 102    | smith    | 11142.8571       | 11142              |
| 103    | allision | 11714.2857       | 11714              |
| 104    | chandler | 7857.1429        | 7857               |
| 105    | charlie  | 12857.1429       | 12857              |
+-----+-----+-----+-----+-----+

```

4.ABS()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES  
(101, 'john', 'ABC@exp.com', '2023-01-15', 65000),  
(102, 'smith', 'jNC@exp.com', '2022-05-20', 78000),  
(103, 'allision', 'MNC@exp.com', '2021-11-10', 82000),  
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT  
    emp_id,  
    salary,  
    (salary - 70000) AS difference_from_avg,  
    ABS(salary - 70000) AS absolute_difference  
FROM employee;
```

Output:

```
+-----+  
| Tables_in_sandbox_db |  
+-----+  
| employee              |  
+-----+
```

```
+-----+-----+-----+-----+-----+  
| emp_id | emp_name | email      | joining_date | salary |  
+-----+-----+-----+-----+-----+  
| 101    | john     | ABC@exp.com | 2023-01-15   | 65000 |  
| 102    | smith    | jNC@exp.com | 2022-05-20   | 78000 |  
| 103    | allision | MNC@exp.com | 2021-11-10   | 82000 |  
| 104    | chandler | YUO@exp.com | 2023-08-01   | 55000 |  
| 105    | charlie  | GHe@exp.com | 2020-03-12   | 90000 |  
+-----+-----+-----+-----+-----+  
+-----+-----+-----+-----+-----+  
| emp_id | salary | difference_from_avg | absolute_difference |  
+-----+-----+-----+-----+-----+  
| 101    | 65000 | -5000              | 5000                |  
| 102    | 78000 | 8000               | 8000                |  
| 103    | 82000 | 12000              | 12000               |  
| 104    | 55000 | -15000             | 15000               |  
| 105    | 90000 | 20000              | 20000               |  
+-----+-----+-----+-----+-----+
```

5.MOD()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    salary INT  
);  
  
INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES  
(101, 'john', 'ABC@exp.com', '2023-01-15', 65000),  
(102, 'smith', 'jnc@exp.com', '2022-05-20', 78000),  
(103, 'allision', 'MNC@exp.com', '2021-11-10', 82000),  
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', 90000);  
  
SHOW TABLES;  
  
SELECT * FROM employee;  
  
SELECT  
    emp_id,  
    salary,  
    MOD(salary, 10000) AS salary_remainder  
FROM employee;;
```

Output:

```
+-----+  
| Tables_in_sandbox_db |  
+-----+  
| employee |  
+-----+  
  
+-----+-----+-----+-----+-----+  
| emp_id | emp_name | email | joining_date | salary |  
+-----+-----+-----+-----+-----+  
| 101 | john | ABC@exp.com | 2023-01-15 | 65000 |  
| 102 | smith | jnc@exp.com | 2022-05-20 | 78000 |  
| 103 | allision | MNC@exp.com | 2021-11-10 | 82000 |  
| 104 | chandler | YUO@exp.com | 2023-08-01 | 55000 |  
| 105 | charlie | GHe@exp.com | 2020-03-12 | 90000 |  
+-----+-----+-----+-----+-----+  
  
+-----+-----+-----+  
| emp_id | salary | salary_remainder |  
+-----+-----+-----+  
| 101 | 65000 | 5000 |  
| 102 | 78000 | 8000 |  
| 103 | 82000 | 2000 |  
| 104 | 55000 | 5000 |  
| 105 | 90000 | 0 |  
+-----+-----+-----+
```

6.POWER()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, salary) VALUES  
(101, 'john', 'ABC@exp.com', '2023-01-15', 65000),  
(102, 'smith', 'JNC@exp.com', '2022-05-20', 78000),  
(103, 'allision', 'MNC@exp.com', '2021-11-10', 82000),  
(104, 'chandler', 'YUO@exp.com', '2023-08-01', 55000),  
(105, 'charlie', 'GHE@exp.com', '2020-03-12', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT  
    emp_id,  
    emp_name,  
    POWER(emp_id, 2) AS emp_id_squared  
FROM employee;
```

Output:

```
+-----+  
| Tables_in_sandbox_db |  
+-----+  
| employee              |  
+-----+  
  
+-----+-----+-----+-----+-----+  
| emp_id | emp_name | email      | joining_date | salary |  
+-----+-----+-----+-----+-----+  
| 101    | john     | ABC@exp.com | 2023-01-15   | 65000 |  
| 102    | smith    | JNC@exp.com | 2022-05-20   | 78000 |  
| 103    | allision | MNC@exp.com | 2021-11-10   | 82000 |  
| 104    | chandler | YUO@exp.com | 2023-08-01   | 55000 |  
| 105    | charlie  | GHE@exp.com | 2020-03-12   | 90000 |  
+-----+-----+-----+-----+-----+  
  
+-----+-----+-----+  
| emp_id | emp_name | emp_id_squared |  
+-----+-----+-----+  
| 101    | john     | 10201          |  
| 102    | smith    | 10404          |  
| 103    | allision | 10609          |  
| 104    | chandler | 10816          |  
| 105    | charlie  | 11025          |  
+-----+-----+-----+
```

Date functions

3.DATE FUNCTIONS:

1.YEAR()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    dob DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date,dob, salary) VALUES  
(101, ' john ', 'ABC@exp.com', '2023-01-15','1995-04-12', 65000),  
(102, ' smith', 'jnc@exp.com', '2022-05-20','1990-08-25', 78000),  
(103, ' allision ', 'MNC@exp.com', '2021-11-10','1988-12-03', 82000),  
(104, 'chandler', 'YUO@exp.com', '2023-08-01','1998-02-14', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12','1992-06-30', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT  
    emp_id,  
    joining_date,  
    YEAR(joining_date) AS joining_year  
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | dob      | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith     | jnc@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler  | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | joining_date | joining_year |
+-----+-----+-----+
| 101    | 2023-01-15   | 2023         |
| 102    | 2022-05-20   | 2022         |
| 103    | 2021-11-10   | 2021         |
| 104    | 2023-08-01   | 2023         |
| 105    | 2020-03-12   | 2020         |
+-----+-----+-----+
```

2.MONTH()-

```
CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    email VARCHAR(100),
    joining_date DATE,
    dob DATE,
    salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date,dob, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15','1995-04-12', 65000),
(102, ' smith', 'jnc@exp.com', '2022-05-20','1990-08-25', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10','1988-12-03', 82000),
(104, 'chandler', 'YUQ@exp.com', '2023-08-01','1998-02-14', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12','1992-06-30', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
    emp_id,
    joining_date,
    MONTH(joining_date) AS joining_month
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | dob      | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john    | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith   | jnc@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | joining_date | joining_month |
+-----+-----+-----+
| 101    | 2023-01-15   | 1             |
| 102    | 2022-05-20   | 5             |
| 103    | 2021-11-10   | 11            |
| 104    | 2023-08-01   | 8             |
| 105    | 2020-03-12   | 3             |
+-----+-----+-----+
```

3.DAY()-

```
CREATE TABLE employee (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  email VARCHAR(100),
  joining_date DATE,
  dob DATE,
  salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date,dob, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15','1995-04-12', 65000),
(102, ' smith', 'jnc@exp.com', '2022-05-20','1990-08-25', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10','1988-12-03', 82000),
(104, 'chandler', 'YUQ@exp.com', '2023-08-01','1998-02-14', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12','1992-06-30', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
  emp_id,
  joining_date,
  DAY(joining_date) AS joining_day
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | dob       | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith     | jNC@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler  | YUO@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | joining_date | joining_day |
+-----+-----+-----+
| 101    | 2023-01-15   | 15          |
| 102    | 2022-05-20   | 20          |
| 103    | 2021-11-10   | 10          |
| 104    | 2023-08-01   | 1           |
| 105    | 2020-03-12   | 12          |
+-----+-----+-----+
```

4.TIMESTAMPDIFF()-

```
CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    email VARCHAR(100),
    joining_date DATE,
    dob DATE,
    salary INT
);

INSERT INTO employee (emp_id, emp_name, email, joining_date,dob, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15','1995-04-12', 65000),
(102, ' smith', 'jNC@exp.com', '2022-05-20','1990-08-25', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10','1988-12-03', 82000),
(104, 'chandler', 'YUO@exp.com', '2023-08-01','1998-02-14', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12','1992-06-30', 90000);

SHOW TABLES;

SELECT * FROM employee;

SELECT
    emp_id,
    emp_name,
    dob,
    TIMESTAMPDIFF(YEAR, dob, CURDATE()) AS age_in_years,
    joining_date,
    TIMESTAMPDIFF(MONTH, joining_date, CURDATE()) AS experience_in_months
FROM employee;
```

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email | joining_date | dob | salary |
+-----+-----+-----+-----+-----+-----+
| 101 | john | ABC@exp.com | 2023-01-15 | 1995-04-12 | 65000 |
| 102 | smith | jNC@exp.com | 2022-05-20 | 1990-08-25 | 78000 |
| 103 | allision | MNC@exp.com | 2021-11-10 | 1988-12-03 | 82000 |
| 104 | chandler | YUO@exp.com | 2023-08-01 | 1998-02-14 | 55000 |
| 105 | charlie | GHe@exp.com | 2020-03-12 | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dob | age_in_years | joining_date | experience_in_months |
+-----+-----+-----+-----+-----+-----+
| 101 | john | 1995-04-12 | 31 | 2023-01-15 | 43 |
| 102 | smith | 1990-08-25 | 36 | 2022-05-20 | 51 |
| 103 | allision | 1988-12-03 | 37 | 2021-11-10 | 57 |
| 104 | chandler | 1998-02-14 | 28 | 2023-08-01 | 36 |
| 105 | charlie | 1992-06-30 | 34 | 2020-03-12 | 77 |
+-----+-----+-----+-----+-----+-----+
```


Aggregate functions

4.AGGREGATE FUNCTIONS:

1.COUNT()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    dob DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES  
(101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),  
(102, 'smith', 'jNC@exp.com', '2022-05-20', '1990-08-25', 78000),  
(103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),  
(104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT
```

```
    COUNT(emp_id) AS total_employees
```

```
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name   | email       | joining_date | dob         | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john       | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000  |
| 102    | smith      | jNC@exp.com | 2022-05-20   | 1990-08-25 | 78000  |
| 103    | allision   | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000  |
| 104    | chandler   | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000  |
| 105    | charlie    | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000  |
+-----+-----+-----+-----+-----+-----+

+-----+
| total_employees |
+-----+
|          5      |
+-----+
```

2.SUM()-

```
CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    email VARCHAR(100),
    joining_date DATE,
    dob DATE,
    salary INT
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES
(101, ' john ', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),
(102, ' smith', 'jNC@exp.com', '2022-05-20', '1990-08-25', 78000),
(103, ' allision ', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),
(104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),
(105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT
```

```
    SUM(salary) AS total_company_payroll
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email | joining_date | dob | salary |
+-----+-----+-----+-----+-----+-----+
| 101 | john | ABC@exp.com | 2023-01-15 | 1995-04-12 | 65000 |
| 102 | smith | jNC@exp.com | 2022-05-20 | 1990-08-25 | 78000 |
| 103 | allision | MNC@exp.com | 2021-11-10 | 1988-12-03 | 82000 |
| 104 | chandler | YUQ@exp.com | 2023-08-01 | 1998-02-14 | 55000 |
| 105 | charlie | GHe@exp.com | 2020-03-12 | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+
| total_company_payroll |
+-----+
| 370000 |
+-----+
```

3.AVG()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    dob DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES  
(101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),  
(102, 'smith', 'jNC@exp.com', '2022-05-20', '1990-08-25', 78000),  
(103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),  
(104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT
```

```
    ROUND(AVG(salary)) AS average_employee_salary  
FROM employee;
```

Output:

+-----+						
Tables_in_sandbox_db						
+-----+						
employee						
+-----+						
+-----+-----+-----+-----+-----+-----+						
emp_id	emp_name	email	joining_date	dob	salary	
+-----+-----+-----+-----+-----+-----+						
101	john	ABC@exp.com	2023-01-15	1995-04-12	65000	
102	smith	jNC@exp.com	2022-05-20	1990-08-25	78000	
103	allision	MNC@exp.com	2021-11-10	1988-12-03	82000	
104	chandler	YUQ@exp.com	2023-08-01	1998-02-14	55000	
105	charlie	GHe@exp.com	2020-03-12	1992-06-30	90000	
+-----+-----+-----+-----+-----+-----+						
+-----+						
average_employee_salary						
+-----+						
74000						
+-----+						

4.MIN() AND MAX()-

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    email VARCHAR(100),  
    joining_date DATE,  
    dob DATE,  
    salary INT  
);
```

```
INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES  
(101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),  
(102, 'smith', 'jNC@exp.com', '2022-05-20', '1990-08-25', 78000),  
(103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),  
(104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),  
(105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
```

```
SHOW TABLES;
```

```
SELECT * FROM employee;
```

```
SELECT  
    MIN(salary) AS lowest_salary,  
    MAX(salary) AS highest_salary,  
    MIN(joining_date) AS earliest_join_date,  
    MAX(joining_date) AS new_join_date  
FROM employee;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | dob      | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john     | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith    | jNC@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie  | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+-----+
| lowest_salary | highest_salary | earliest_join_date | new_join_date |
+-----+-----+-----+-----+-----+-----+
| 55000         | 90000          | 2020-03-12         | 2023-08-01    |
+-----+-----+-----+-----+-----+-----+
```

5.GROUPBY()-

```
1  CREATE TABLE employee (
2      emp_id INT PRIMARY KEY,
3      emp_name VARCHAR(50),
4      email VARCHAR(100),
5      joining_date DATE,
6      dob DATE,
7      salary INT
8  );
9
10 INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES
11 (101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),
12 (102, 'smith', 'jNC@exp.com', '2022-05-20', '1990-08-25', 78000),
13 (103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),
14 (104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),
15 (105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
16
17 SHOW TABLES;
18
19 SELECT * FROM employee;
20
21
22 SELECT
23     YEAR(joining_date) AS joining_year,
24     COUNT(emp_id) AS total_employees,
25     SUM(salary) AS total_salary
26 FROM employee
27 GROUP BY YEAR(joining_date);
28
```


Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | email      | joining_date | dob      | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john    | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith   | jnc@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie | GHE@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| joining_year | total_employees | total_salary |
+-----+-----+-----+
| 2023         | 2               | 120000       |
| 2022         | 1               | 78000        |
| 2021         | 1               | 82000        |
| 2020         | 1               | 90000        |
+-----+-----+-----+
```

6.HAVING()-

```
1 CREATE TABLE employee (
2     emp_id INT PRIMARY KEY,
3     emp_name VARCHAR(50),
4     email VARCHAR(100),
5     joining_date DATE,
6     dob DATE,
7     salary INT
8 );
9
10 INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES
11 (101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),
12 (102, 'smith', 'jnc@exp.com', '2022-05-20', '1990-08-25', 78000),
13 (103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),
14 (104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),
15 (105, 'charlie', 'GHE@exp.com', '2020-03-12', '1992-06-30', 90000);
16
17 SHOW TABLES;
18
19 SELECT * FROM employee;
20
21
22
23 SELECT
24     YEAR(joining_date) AS joining_year,
25     AVG(salary) AS avg_salary
26 FROM employee
27 GROUP BY YEAR(joining_date)
28 HAVING AVG(salary) > 70000;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | dob       | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000 |
| 102    | smith     | jnc@exp.com | 2022-05-20   | 1990-08-25 | 78000 |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000 |
| 104    | chandler  | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000 |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000 |
+-----+-----+-----+-----+-----+-----+

+-----+-----+
| joining_year | avg_salary |
+-----+-----+
| 2022         | 78000.0000 |
| 2021         | 82000.0000 |
| 2020         | 90000.0000 |
+-----+-----+
```

7.ORDERBY()-

```
1  CREATE TABLE employee (
2      emp_id INT PRIMARY KEY,
3      emp_name VARCHAR(50),
4      email VARCHAR(100),
5      joining_date DATE,
6      dob DATE,
7      salary INT
8  );
9
10 INSERT INTO employee (emp_id, emp_name, email, joining_date, dob, salary) VALUES
11 (101, 'john', 'ABC@exp.com', '2023-01-15', '1995-04-12', 65000),
12 (102, 'smith', 'jnc@exp.com', '2022-05-20', '1990-08-25', 78000),
13 (103, 'allision', 'MNC@exp.com', '2021-11-10', '1988-12-03', 82000),
14 (104, 'chandler', 'YUQ@exp.com', '2023-08-01', '1998-02-14', 55000),
15 (105, 'charlie', 'GHe@exp.com', '2020-03-12', '1992-06-30', 90000);
16
17 SHOW TABLES;
18
19 SELECT * FROM employee;
20
21 SELECT
22     emp_id,
23     emp_name,
24     salary
25 FROM employee
26 ORDER BY salary DESC;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employee              |
+-----+

+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name  | email      | joining_date | dob        | salary |
+-----+-----+-----+-----+-----+-----+
| 101    | john      | ABC@exp.com | 2023-01-15   | 1995-04-12 | 65000  |
| 102    | smith     | jNC@exp.com | 2022-05-20   | 1990-08-25 | 78000  |
| 103    | allision  | MNC@exp.com | 2021-11-10   | 1988-12-03 | 82000  |
| 104    | chandler  | YUQ@exp.com | 2023-08-01   | 1998-02-14 | 55000  |
| 105    | charlie   | GHe@exp.com | 2020-03-12   | 1992-06-30 | 90000  |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| emp_id | emp_name  | salary |
+-----+-----+-----+
| 105    | charlie   | 90000  |
| 103    | allision  | 82000  |
| 102    | smith     | 78000  |
| 101    | john      | 65000  |
| 104    | chandler  | 55000  |
+-----+-----+-----+
```
